TEXT TO SQL PROJECT

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LLM USED CHATGPT AND VERSION 3.5

GRADING

- The grading methodology would be, for each question 5 points would be awarded for a question correct and 5 points taken way for a question being wrong.
- Points off for each time it has to be corrected
- loss of points for if it not working or if it doesn't give you the desired answer
- Points for simplicity

QUESTION 1

Prompt :

--1. Display the cities that makes the most different kinds of products. Experiment with the rank() function.

CODE

- SELECT
- city,
- COUNT(DISTINCT prodId) AS num_products
- FROM
- Products
- GROUP BY
- city
- ORDER BY
- COUNT(DISTINCT prodId) DESC
- LIMIT 1;

GRADE AND FIX

- 4/5
- Only needed 1 fix and that was to simplify the code.

RESULTS OF THE CODE

- The results between my code and the llm where the same.

	city text	num_products bigint
1	Newark	3

QUESTION 2

- --2. Display the names of products whose priceUSD is less than 1% of the average priceUSD, in alphabetical order. from A to Z.

CODE

- SELECT
- name
- FROM
- Products
- WHERE
- priceUSD < (SELECT AVG(priceUSD) * 0.01 FROM Products)
- ORDER BY
- name ASC;

GRADE AND FIX

- 4/5
- Only needed 1 fix and that was to simplify the code.

RESULTS OF THE CODE

- The results between my code and the llm where the same.

	name text
1	Denis Wick Valve Oil
2	Flux Capacitor
3	HAL 9000 memory chip
4	LCARS module

QUESTION 3

- --3. Display the customer last name, product id ordered, and the totalUSD for all orders made in March of any year, sorted by totalUSD from low to high.

CODE

```
SELECT
    p.lastName AS customer_last_name,
    o.prodId AS product_id_ordered,
    o.totalUSD
FROM
    Orders o
JOIN
    Customers c ON o.custId = c.pid
JOIN
    People p ON c.pid = p.pid
WHERE
    EXTRACT(MONTH FROM o.dateOrdered) = 3
ORDER BY
    o.totalUSD;
```

GRADE AND FIX

- 4/5
- Only needed 1 fix and that was to simplify the code.

RESULTS OF THE CODE

- The results between my code and the llm where the same.

	customer_last_name text	product_id_ordered character	totalusd numeric (12,2)
1	Ferguson	p05	6550.98
2	Ferguson	p06	31236.48
3	Armstrong	p01	56671.55

QUESTION 4

--4. Display the last name of all customers (in reverse alphabetical order) and their total ordered by customer, and nothing more.

- --Use coalesce to avoid showing NULL totals.

CODE

```
SELECT
    p.lastName AS customer_last_name,
    COALESCE(SUM(o.totalUSD), 0) AS total_ordered
FROM
    Customers c
LEFT JOIN
    Orders o ON c.pid = o.custId
LEFT JOIN
    People p ON c.pid = p.pid
GROUP BY
    p.lastName
ORDER BY
    p.lastName DESC;
```

GRADE AND FIX

- 3/5
- The first time it showed all people not just customers. So I told it to just show only customers not people.

RESULTS OF THE CODE OF THE LLM

- The final results between the llm and my code where the same.

	customer_last_name text	total_ordered numeric
1	Severinsen	142768.32
2	Robinson	0
3	Morrison	109053.62
4	Ferguson	146525.21
5	Armstrong	72442.75

QUESTION 5

--5. Display the names of all customers who bought products from agents based in Chilliwack along with the names of the products they ordered,

-- and the names of the agents who sold it to them.

CODE

```
SELECT
    p.lastName AS customer_name,
    pr.name AS product_name,
    a.firstName || ' ' || a.lastName AS agent_name
FROM
    Orders o
JOIN
    Customers c ON o.custId = c.pid
JOIN
    People p ON c.pid = p.pid
JOIN
    Products pr ON o.prodId = pr.prodId
JOIN
    Agents ag ON o.agentId = ag.pid
JOIN
    People a ON ag.pid = a.pid
WHERE
    a.homeCity = 'Chilliwack';
```

GRADE AND FIX

- 4/5
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QUESTION 6

- --6. Write a query to check the accuracy of the totalUSD column in the Orders table.
- --This means calculating Orders.totalUSD from data in other tables and comparing those values to the values in Orders.totalUSD.
- --Display all rows in Orders where Orders.totalUSD is incorrect, if any. If there are any incorrect values, explain why they are wrong.
- --Round to exactly two decimal places.

CODE

```
SELECT
    o.*,
    ROUND(SUM(pr.priceUSD * o.quantityOrdered), 2) AS calculated_total
FROM
    Orders o
JOIN
    Products pr ON o.prodId = pr.prodId
GROUP BY
    o.orderNum
HAVING
    ROUND(SUM(pr.priceUSD * o.quantityOrdered), 2) != o.totalUSD;
```

GRADE AND FIX

- 4/5
- Gave pointless info.

RESULTS OF THE CODE OF THE LLM

- The final results between the llm and my code where the same.
- Apart from the extra code and
- Info

	ordernum [PK] integer	dateordered date	custid integer	agentid integer	prodid character	quantityordered integer	totalusd numeric (12,2)	calculated_total numeric
1	1018	2023-02-14	1	3	p04	600	8050.49	10206.00
2	1025	2022-04-01	8	3	p07	888	799.11	888.00
3	1022	2023-03-15	1	3	p06	450	31236.48	39600.00
4	1016	2021-01-23	8	3	p01	1000	60977.22	67760.00
5	1015	2022-01-23	5	3	p05	1000	15771.20	16610.00
6	1021	2023-02-14	4	5	p01	1000	66086.33	67760.00
7	1024	2023-03-15	5	2	p01	880	56671.55	59628.80
8	1012	2023-01-23	4	3	p03	1200	76096.81	78024.00
9	1020	2023-02-14	4	5	p07	600	585.18	600.00
10	1017	2023-02-14	1	3	p03	500	25643.98	32510.00
11	1019	2023-02-14	1	2	p02	400	16249.28	20600.00
12	1011	2024-01-22	1	2	p01	1100	58794.00	74536.00
13	1023	2023-03-15	1	2	p05	500	6550.98	8305.00
14	1026	2022-05-04	8	5	p03	808	47277.29	52536.16

QUESTIONS 7

- --7. Display the first and last name of all customers who are also agents.

CODE

```
SELECT
    c.firstName AS first_name,
    c.lastName AS last_name
FROM
    People c
JOIN
    Customers cu ON c.pid = cu.pid
JOIN
    Agents a ON c.pid = a.pid;
```

GRADE AND FIX

- 5/5
- It was spotless.

RESULTS OF THE CODE OF THE LLM

- The final results between the llm and my code where the same.

	firstname text	lastname text
1	Louis	Armstrong

REFERENCES

- ChatGPT

OPINIONS

The LLM worked for the most part In answering the questions. It worked well and was smooth and seamless. The only drawback to using it was that I needed to simplify the answers it was giving me. The overall grade would be an 80 overall.